95 investigate, recognise and use the expression $\varepsilon = -\,\mathrm{d}(N\Phi)/\mathrm{d}t$ and explain how it is a consequence of Faraday's and Lenz's laws	

4.5 Particle physics

This topic covers atomic structure, particle accelerators, and the standard quark-lepton model, enabling students to describe the behaviour of matter on a subatomic scale.

This topic is the subject of current research, involving the acceleration and detection of high-energy particles. It may be taught by exploring a range of experiments:

- alpha scattering and the nuclear model of the atom
- accelerating particles to high energies
- detecting and interpreting interactions between particles.

Students will be assessed on their ability to:	Suggested experiments
96 use the terms nucleon number (mass number) and proton number (atomic number)	
97 describe how large-angle alpha particle scattering gives evidence for a nuclear atom	
98 recall that electrons are released in the process of thermionic emission and explain how they can be accelerated by electric and magnetic fields	
99 explain the role of electric and magnetic fields in particle accelerators (linac and cyclotron) and detectors (general principles of ionisation and deflection only)	
100 recognise and use the expression $r = p/BQ$ for a charged particle in a magnetic field	
101 recall and use the fact that charge, energy and momentum are always conserved in interactions between particles and hence interpret records of particle tracks	

Students will be assessed on their ability to:	Suggested experiments
102 explain why high energies are required to break particles into their constituents and to see fine structure	
103 recognise and use the expression $\Delta E = c^2\Delta m$ in situations involving the creation and annihilation of matter and antimatter particles	
104 use the non-SI units MeV and GeV (energy) and MeV/c^2, GeV/c^2 (mass) and atomic mass unit u, and convert between these and SI units	
105 be aware of relativistic effects and that these need to be taken into account at speeds near that of light (use of relativistic equations not required)	
106 recall that in the standard quark-lepton model each particle has a corresponding antiparticle, that baryons (eg neutrons and protons) are made from three quarks, and mesons (eg pions) from a quark and an antiquark, and that the symmetry of the model predicted the top and bottom quark	
107 write and interpret equations using standard nuclear notation and standard particle symbols (eg π^+, e^-)	
108 use de Broglie's wave equation $\lambda = h/p$	